SCIENCE & TECHNOLOGY PART 1 | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 9: Carbon Compounds

MODEL ANSWER SHEET / ANSWER KEY

118	9	24
QUESTIONS	IN TEXTBOOK	DIAGRAMS

HOW TO USE
Cover the answer box ✓ and answer first. A **TEXTBOOK** *strip means the same question is in your Balbharati Science-1 textbook, at the set and page shown. Tick ☐ each section once revised.*

01	02	03	04	05	06	07
MCQs 12 Q	Correlation 6 Q	Odd One Out 5 Q	One Sentence 22 Q	Match Columns 3 Q	Right or Wrong 15 Q	Explanation 3 Q
08	**09**	**10**	**11**	**12**	**13**	
Scientific Reasons 6 Q	Solve 16 Q	Distinguish 3 Q	Short Notes 5 Q	Answer the Following 20 Q	Answer the Questions 2 Q	

01 Choose the correct option.

12 QUESTIONS ☐ Revised

Q1 Generally, the melting and boiling point of carbon compounds are found to be less than _____ °C.

(a) **300** (b) 100 (c) 200 (d) 150

✓ **ANSWER / SOLUTION**
Generally, the melting and boiling point of carbon compounds are found to be less than **300** °C.

Q2 Number of valence electrons in a carbon atom is _____.

(a) **4** (b) 5 (c) 1 (d) 3

✓ **ANSWER / SOLUTION**
Number of valence electrons in a carbon atom is **4**.

Q3 The bond between two oxygen atoms is _____ bond.

(a) **double** (b) triple
(c) single (d) none of these

✓ **ANSWER / SOLUTION**
The bond between two oxygen atoms is **double** bond .

Q4 The molecular masses of a carbon compound spread over a range of _____.

(a) 10^{12} (b) 10^{14}
(c) 10^{10} (d) 10^{13}

✓ **ANSWER / SOLUTION**
The molecular masses of a carbon compound spread over a range of 10^{12}.

Q5 The unsaturated hydrocarbons containing a carbon-carbon double bond are called _____.

(a) **Alkenes** (b) Alkanes (c) Alkynes
(d) Alcohol

✓ **ANSWER / SOLUTION**
The unsaturated hydrocarbons containing a carbon-carbon double bond are called **Alkenes**.

Q6 The phenomenon in which compounds having different structural formulae have the same molecular formula is called _____.

(a) **structural isomerism** (b) catenation
(c) homologous (d) functional group

✓ **ANSWER / SOLUTION**
The phenomenon in which compounds having different structural formulae have the same molecular formula is called **structural isomerism**.

Q7 From the following hydrocarbon _____ is the cyclic hydrocarbon.

(a) isobutane (b) propyne
(c) **benzene** (d) isobutylene

ANSWER / SOLUTION
From the following hydrocarbon **benzene** is the cyclic hydrocarbon.

Q8 While going in an increasing order there is a rise in the molecular mass of the consecutive members of the homologous series by _____.

(a) $14\ \mathrm{u}$ (b) $15\ \mathrm{u}$
(c) $16\ \mathrm{u}$ (d) $17\ \mathrm{u}$

ANSWER / SOLUTION
While going in an increasing order there is a rise in the molecular mass of the consecutive members of the homologous series by $14\ \mathrm{u}$.

Q9 The general molecular formula for the homologous series of alkynes is _____.

(a) $\mathrm{C}_n\mathrm{H}_{2n}$ (b) $\mathrm{C}_n\mathrm{H}_{2n+2}$
(c) $\mathrm{C}_n\mathrm{H}_{2n-2}$ (d) $\mathrm{C}_n\mathrm{H}_{2n-1}$

ANSWER / SOLUTION
The general molecular formula for the homologous series of alkynes is $\mathrm{C}_n\mathrm{H}_{2n-2}$.

Q10 _____ is one of the combustible components of L.P.G.

(a) Methane (b) Ethane
(c) **Propane** (d) Butanol
(e) Ethene

ANSWER / SOLUTION
Propane is one of the combustible components of L.P.G.

Explanation:
LPG (Liquefied Petroleum Gas) mainly contains propane $\mathrm{C_3H_8}$ and butane $\mathrm{C_4H_{10}}$, which are highly flammable and used as fuels for cooking and heating. Methane and ethane are found in natural gas, while butanol is an alcohol, not a major LPG component.

Q11 At room temperature ethanol is _____.

(a) solid (b) gas (c) plasma (d) **liquid**

ANSWER / SOLUTION
At room temperature ethanol is **liquid**.

Q12 Generally _____ is called spirit.

(a) methanol (b) **ethanol** (c) propanol
(d) butanol

ANSWER / SOLUTION
Generally **ethanol** is called spirit.

02 Find the correlation.

6 QUESTIONS ☐ Revised

Q1 Solid : iodine : : _____ : bromine

ANSWER / SOLUTION
Solid : iodine : : **liquid** : bromine

Q2 $\mathrm{CH_3}$--$\mathrm{CH_2}$--CHO : propanal : : $\mathrm{CH_3}$--COOH : _____

ANSWER / SOLUTION
$\mathrm{CH_3}$--$\mathrm{CH_2}$--CHO : propanal : : $\mathrm{CH_3}$--COOH : **ethanoic acid**

Q3 Ketone : --CO-- : : Ester : _____

ANSWER / SOLUTION
Ketone : --CO-- : : Ester : --COO--

Q4 Cyclohexane : Cyclic hydrocarbon : : Isobutylene : _____

ANSWER / SOLUTION
Cyclohexane : Cyclic hydrocarbon : : Isobutylene : **Branched hydrocarbon**

Q5 Saturated hydrocarbon : Single bond : : Unsaturated hydrocarbon : _____

ANSWER / SOLUTION
Saturated hydrocarbon : Single bond : : Unsaturated hydrocarbon : **Double and/or triple bond**

Q6 Saturated carbon compounds : blue flame : : Unsaturated carbon compounds : _____

ANSWER / SOLUTION
Saturated carbon compounds : blue flame : : Unsaturated carbon compounds : **Yellow flame**

03 Find odd one out.

5 QUESTIONS ☐ Revised

Q1 **Find the odd one out and give its explanation.**

(a) Ethylene (b) Styrene (c) Propylene
(d) **Teflon**

✓ **ANSWER / SOLUTION**

Teflon

Explanation-

Teflon is a polymer while others are monomer units of respective polymers.

Q2 **Find the odd one out and give its explanation.**

(a) Butane (b) Methane (c) Benzene
(d) **Ozone**

✓ **ANSWER / SOLUTION**

Ozone

Explanation-

Ozone is inorganic while others are organic compounds.

Q3 **Find the odd one out and give its explanation.**

(a) CH_4 (b) C_2H_6
(c) C_5H_{12} (d) $CaCO_3$

✓ **ANSWER / SOLUTION**

$CaCO_3$

Explanation-

Calcium carbonate ($CaCO_3$) is an inorganic compound while others are organic compounds.

Q4 **Find the odd one out and give its explanation.**

(a) C_2H_2 (b) C_3H_8
(c) C_2H_6 (d) CH_4

✓ **ANSWER / SOLUTION**

C_2H_2

Explanation-

Ethyne (C_2H_2) is an unsaturated hydrocarbon while others are saturated hydrocarbons.

Q5 **Find the odd one out and give its explanation.**

(a) C_2H_4 (b) C_4H_{10}
(c) C_3H_8 (d) CH_4

✓ **ANSWER / SOLUTION**

C_2H_4

Explanation-

Ethene (C_2H_4) is an unsaturated hydrocarbon while others are saturated hydrocarbons.

04 Answer in one sentence.

22 QUESTIONS ☐ Revised

Q1 **Write the molecular formula of the given compound.**
Ethyl ethanoate

✓ **ANSWER / SOLUTION**

Ethyl ethanoate- $C_4H_8O_2$

Q2 **Write the molecular formula of the given compound.**
Sodium ethanoate

✓ **ANSWER / SOLUTION**

Sodium ethanoate- $C_2H_3O_2Na$

Q3 **Write the molecular formula of the given compound.**
Sodium ethoxide

✓ **ANSWER / SOLUTION**

Sodium ethoxide- C_2H_5ONa

Q4 **Write the molecular formula of the given compound.**
Stearic acid

✓ **ANSWER / SOLUTION**

Stearic acid- $C_{17}H_{35}COOH$ **or** $C_{18}H_{36}O_2$

Q5 **Write the molecular formula of the given compound.**
Oleic acid

ANSWER / SOLUTION
Oleic acid- $C_{17}H_{33}COOH$ **or** $C_{18}H_{34}O_2$

Q6 **Write the molecular formula of the given compound.**
Palmitic acid

ANSWER / SOLUTION
Palmitic acid- $C_{15}H_{31}COOH$ **or** $C_{16}H_{32}O_2$

Q7 **Write the molecular formula of the given compound.**
Ethylene

ANSWER / SOLUTION
Ethylene- C_2H_4

Q8 **Write the molecular formula of the given compound.**
Benzene

ANSWER / SOLUTION
Benzene- C_6H_6

Q9 **Write the molecular formula of the given compound.**
Acetic acid

ANSWER / SOLUTION
Acetic acid- CH_3COOH **or** $C_2H_4O_2$

Q10 **Write the molecular formula of the given compound.**
Propylene

ANSWER / SOLUTION
Propylene- C_3H_6

Q11 **Write the molecular formula of the given compound.**
Acetylene

ANSWER / SOLUTION
Acetylene- C_2H_2

Q12 **Write the molecular formula of the given compound.**
Ethyl alcohol

ANSWER / SOLUTION
Ethyl alcohol- C_2H_5OH **or** C_2H_6O

Q13 **Write the molecular formula of the given compound.**
Acetone

ANSWER / SOLUTION
Acetone- C_3H_6O

Q14 **Write the molecular formula of the given compound.**
Propene

ANSWER / SOLUTION
Propene- C_3H_6

Q15 **Write the molecular formula of the given compound.**
Ethanol

ANSWER / SOLUTION
Ethanol- C_2H_5OH **or** C_2H_6O

Q16 **Write the molecular formula of the given compound.**
Ethanoic acid

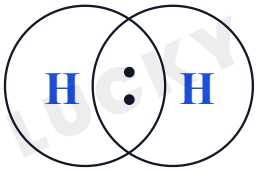
ANSWER / SOLUTION
Ethanoic acid- CH_3COOH **or** $C_2H_4O_2$

Q17 **Write the molecular formula of the given compound.**
Isobutane

ANSWER / SOLUTION
Isobutane- C_4H_{10}

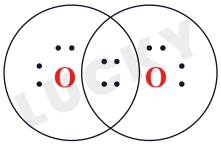
Q18 **Draw electron dot structure and line structure for a given molecule.**
Hydrogen

ANSWER / SOLUTION

Molecule	Electron dot structure	Line structure
Hydrogen (H_2)		$H-H$

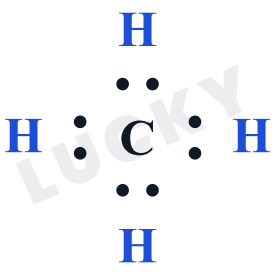
Q19 **Draw electron dot structure and line structure for a given molecule.**
Oxygen

ANSWER / SOLUTION

Molecule	Electron dot structure	Line structure
Oxygen (O_2)		

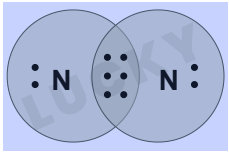
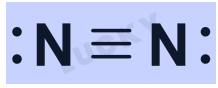
Q20 **Draw electron dot structure and line structure for a given molecule.**
Methane

ANSWER / SOLUTION

Molecule	Electron dot structure	Line structure
Methane (CH_4)		$\begin{array}{c} H \\ \vert \\ H-C-H \\ \vert \\ H \end{array}$

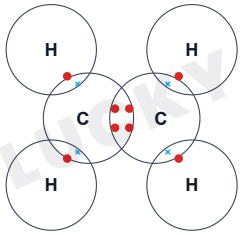
Q21 **Draw electron dot structure and line structure for a given molecule.**
Nitrogen

ANSWER / SOLUTION

Molecule	Electron dot structure	Line structure
Nitrogen (N_2)		

Q22 **Draw electron dot structure and line structure for a given molecule.**
Ethene

ANSWER / SOLUTION

Molecule	Electron dot structure	Line structure
Ethene (C_2H_4)		$CH_2=CH_2$

05 Match the columns

3 QUESTIONS · 1 ☐ Revised

Q1 Match the pairs.

Group 'A'	Group 'B'
a. C_2H_6	1. Unsaturated hydrocarbon
b. C_2H_2	2. Molecular formula of an alcohol
c. CH_4O	3. Saturated hydrocarbon
d. C_3H_6	4. Triple bond

TEXTBOOK Exercise · Q.1 · Page 133

✓ ANSWER / SOLUTION

Group 'A'	Group 'B'
a. C_2H_6	1. Saturated hydrocarbon
b. C_2H_2	2. Triple bond
c. CH_4O	3. Molecular formula of an alcohol
d. C_3H_6	4. Unsaturated hydrocarbon

Q2 Match the columns.

Group A	Group B
1. Straight chain hydrocarbon	a) Benzene
2. Branched chain hydrocarbon	b) Propyne
3. Cyclic hydrocarbon	c) Isobutylene

✓ ANSWER / SOLUTION

Group A	Group B
1. Straight chain hydrocarbon	b) Propyne
2. Branched chain hydrocarbon	c) Isobutylene
3. Cyclic hydrocarbon	a) Benzene

Q3 Match the columns.

Group A	Group B
1) Ether	a) --OH
2) Ketone	b) --O--
3) Ester	c) --CO--
4) Alcohol	d) --COO--

✓ ANSWER / SOLUTION

Group A	Group B
1) Ether	b) --O--
2) Ketone	c) --CO--
3) Ester	d) --COO--
4) Alcohol	a) --OH

06 Right or Wrong sentence

15 QUESTIONS ☐ Revised

Q1 The number of electrons in the valence shell of a carbon atom is 4.

(a) **Right** (b) Wrong

✓ ANSWER / SOLUTION

The number of electrons in the valence shell of a carbon atom is 4- **Right**

Q2 Your body is made up of carbon.

(a) **Right** (b) Wrong

✓ ANSWER / SOLUTION

Your body is made up of carbon- **Right**

Q3 Carbon compounds contain only open chains of carbon atoms.

(a) Right (b) **Wrong**

✓ ANSWER / SOLUTION

Carbon compounds contain only open chains of carbon atoms- **Wrong**

Q4 In general, saturated compounds are more reactive than unsaturated compounds.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

In general, saturated compounds are more reactive than unsaturated compounds- **Wrong**

Q5 Benzene is a cyclic unsaturated hydrocarbon.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Benzene is a cyclic unsaturated hydrocarbon- **Right**

Q6 Cyclohexane is a branched chain type of hydrocarbon.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

Cyclohexane is a branched chain type of hydrocarbon- **Wrong**

Q7 As one ascends in any homologous series, physical properties change gradually.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

As one ascends in any homologous series, physical properties change gradually- **Right**

Q8 There are different general molecular formula for all members of the homologous series.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

There are different general molecular formula for all members of the homologous series- **Wrong**

Q9 In LPG, butane is a flammable component.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

The statement is **Right**.

Explanation:

Liquefied Petroleum Gas (LPG) contains combustible butane and propane. Both extremely combustible gases are used as fuel in homes and factories.

Q10 Substances that can give oxygen to other substances are called reductant.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

The statement is **Wrong**.

Explanation:

Oxidising agents, not reductants, provide oxygen. A reductant provides electrons to another chemical, reducing it while oxidising the reductant.

Q11 Potassium permanganate is an oxidizing compound in regular use.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Potassium permanganate is an oxidizing compound in regular use- **Right**

Q12 Colorless ethanol is in liquid state at room temperature.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Colorless ethanol is in liquid state at room temperature- **Right**

Q13 Ethanol is soluble in water in all proportions.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Ethanol is soluble in water in all proportions- **Right**

Q14 Ester is a sweet-smelling compound.

(a) **Right** (b) Wrong

ANSWER / SOLUTION

Ester is a sweet-smelling compound- **Right**

Q15 Two carbon atoms can always form one or two covalent bonds.

(a) Right (b) **Wrong**

ANSWER / SOLUTION

This statement is **Wrong**.

07 Write an Explanation

3 QUESTIONS · 1 ☐ Revised

Q1 **Write an Explanation.**
Alkane

TEXTBOOK Exercise · Q.4 (e) · Page 133

ANSWER / SOLUTION

The saturated hydrocarbons containing only single bonds between carbon atoms are called alkanes.

Q2 **Write an Explanation.**
Alkene

ANSWER / SOLUTION

An alkene is an unsaturated hydrocarbon that contains at least one carbon-carbon double bond.

Q3 **Write an Explanation.**
Alkyne

ANSWER / SOLUTION

An alkyne is a type of unsaturated hydrocarbon characterized by the presence of at least one carbon-carbon triple bond.

08 Write scientific reasons

6 QUESTIONS ☐ Revised

Q1 **Write scientific reason.**
Ethylene is an unsaturated hydrocarbon.

ANSWER / SOLUTION

1. Hydrocarbons containing at least one carbon-carbon double or triple bond are called unsaturated hydrocarbons.
2. The structure of ethylene is $CH_2 = CH_2$. It has a double bond and hence, it is an unsaturated hydrocarbon.

Q2 **Write scientific reason.**
The flame appears yellow in the ignition of naphthalene.

ANSWER / SOLUTION

1. Naphthalene is an unsaturated hydrocarbon.
2. Unsaturated hydrocarbons burns with a yellow flame and releases black soot/smoke.
   Hence, the flame appears yellow in the ignition of naphthalene.

Q3 **Write scientific reason.**
The color of iodine disappears in the reaction between vegetable oil and tincture iodine.

ANSWER / SOLUTION

1. Vegetable oils are unsaturated fatty acids.
2. They contain carbon-carbon multiple bonds and thus, undergo addition reaction with iodine.
   Hence, the color of iodine disappears in the reaction between vegetable oil and tincture iodine.

Q4 **Write scientific reason.**
Vegetable ghee is formed from the hydrogenation of vegetable oil in presence of nickel catalyst.

ANSWER / SOLUTION

1. Vegetable oil contains carbon-carbon multiple bonds while vanaspati ghee contains carbon-carbon single bonds. So, vegetable oil is an unsatured compound while vanaspati ghee is a saturated compound.
2. An unsaturated compound undergoes addition reaction with hydrogen in presence of nickel catalyst to form a saturated compound. This reaction is called hydrogenation.
   Hence, vanaspati ghee is formed from the hydrogenation of vegetable oil in presence of nickel catalyst.

Q5 **Write scientific reason.**
Carbon has the property of forming many compounds.

ANSWER / SOLUTION

1. Carbon is a tetravalent atom with the property of catenation. Thus, it forms compounds in which the carbon atoms are arranged in the form of straight chains, branched chains or rings.
2. It can form single and multiple covalent bonds with other carbon atoms.
3. Carbon is tetravalent. Thus, it can form four covalent bonds with carbon or other elements like oxygen, nitrogen, sulphur, halogens and phosphorus.
4. One more characteristics of carbon, which is responsible for large number of carbon compounds is isomerism.
   Hence, carbon has the property of forming many compounds.

Q6 **Write scientific reason.**
Benzene compounds are called aromatic compounds.

ANSWER / SOLUTION

1. Benzene compounds are cyclic unsaturated hydrocarbons having structure similar to benzene.
2. There are three alternate double bonds in the six-membered ring structure of benzene.
3. Compounds having this characteristic unit in their structures are called aromatic compounds.
   Hence, benzene compounds are called aromatic compounds.

09 Solve the following Questions

16 QUESTIONS · 7 ☐ Revised

Q1 Explain the Catenation power.

ANSWER / SOLUTION

1. Carbon has a unique ability to form strong covalent bonds with other carbon atoms; this results in the formation of big molecules. This property of carbon is called catenation power.
2. Due to the catenation power of carbon, the compounds formed may have long straight chains of carbon, branched chains of carbon or rings of carbon.

Q2 Explain covalent bond with example.

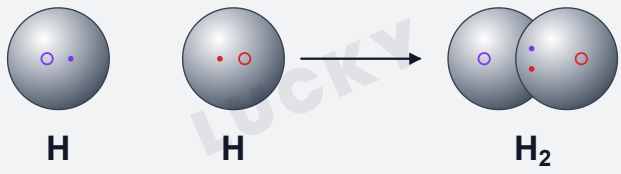
TEXTBOOK **Exercise · Q.4 (b) · Page 133**

ANSWER / SOLUTION

The chemical bond which is formed by sharing of two valence electrons between two atoms is called a covalent bond.

E.g. Formation of H_2 molecule:

The atomic number of hydrogen is 1. It has 1 electron in its valence shell. It requires one more electron to complete its duplet and attain the stable electronic configuration of helium. Therefore, two hydrogen atoms share their electrons with each other to form H_2 molecule. A single covalent bond is formed between two hydrogen atoms by sharing of two electrons (one pair of shared electrons).



OR $H-H$

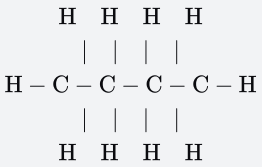
Q3 Explain the Structural isomerism term with example.

TEXTBOOK Exercise · Q.4 (a) · Page 133

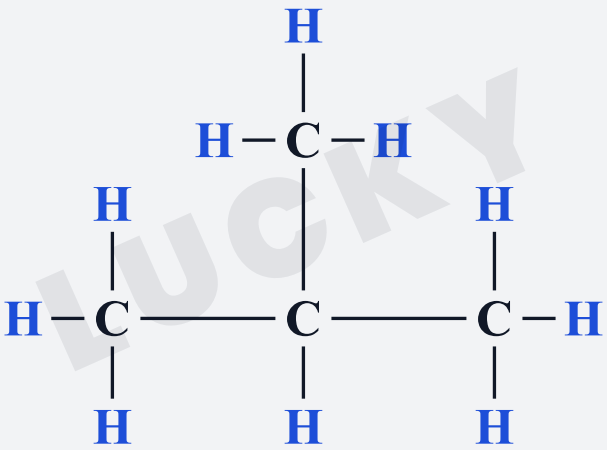
ANSWER / SOLUTION

It is a phenomenon in which the compounds having different structural formulas have the same molecular formula.

E.g. n-Butane and isobutane are structural isomers as they have different structural formulae but have a same molecular formula: C_4H_{10}.



n-Butane (C_4H_{10})



Isobutane (C_4H_{10})

Q4 What is an oxidising agent?

ANSWER / SOLUTION

An oxidising agent is a reactant which readily transfers oxygen atoms to another substance by reducing itself in the process.

Solution 2 (Alternative):

The substances which can give oxygen to other substances are called oxidant or oxidizing agents.

E.g. Potassium permanganate ($KMnO_4$) and potassium dichromate ($K_2Cr_2O_7$) in acidic or alkaline medium is commonly used oxidants.

Q5 Explain the following term with example. Reduction

TEXTBOOK Exercise · Q.4 (i) · Page 133

ANSWER / SOLUTION

A chemical reaction in which a substance gains hydrogen or loses oxygen is called a reduction reaction.

E.g. In the following reaction, ethylene (ethene) is reduced to ethane.

$$H_2C = CH_2 \xrightarrow[Pt/Ni]{H_2} CH_3CH_3$$

Q6 Explain the concept of heteroatoms with the help of examples.

TEXTBOOK Exercise · Q.4 (c) · Page 133

ANSWER / SOLUTION

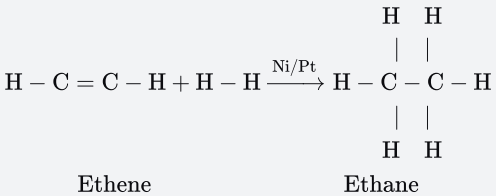
1. Many carbon compounds are formed in which atoms of elements such as oxygen, nitrogen, sulphur, halogens, etc., substitute one or more hydrogen atoms in the hydrocarbon chain.
2. The atom of an element other than carbon and hydrogen which replaces one or more hydrogen atoms in the hydrocarbon chain is called a heteroatom.
3. The compound acquires specific chemical properties due to these hetero atoms or the groups of atoms that contain heteroatoms, irrespective of the length and nature of the carbon chain in that compound. Therefore, these hetero atoms or the groups of atoms containing hetero atoms are called functional groups.
4. **E.g.** In CH_3OH, the functional group contains hetero atom 'O'.
   In CH_3CH_2Cl, the functional group contains hetero atom 'Cl'.

Q7 Explain the following reaction with example. Addition reaction

ANSWER / SOLUTION

The reaction in which the addition of a substance occurs across the carbon-carbon double or triple bond is called an addition reaction. Addition reactions occur in unsaturated hydrocarbons.

e.g.



Ethene Ethane

Q8 **Explain the following reaction with an example.**
Substitution reaction

ANSWER / SOLUTION

The reaction in which replacement of an atom or a group of atoms in a molecule by another atom or group of atoms takes place is called substitution reaction. The substitution reactions of alkanes with chlorine occur in presence of sunlight in which one or more hydrogen atoms of alkanes such as methane are replaced by chlorine atoms.

E.g.

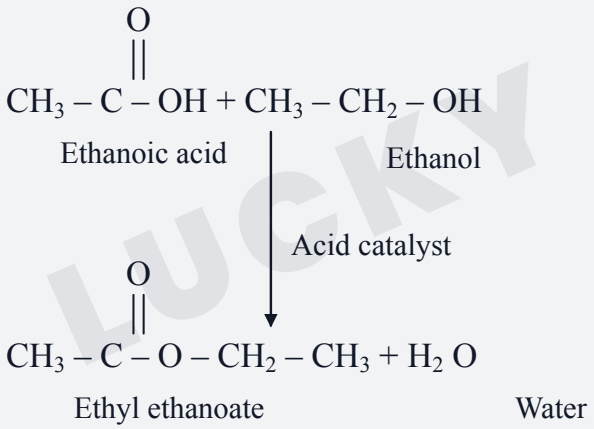
$$CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$$

Q9 **Explain the following reaction with an example.**
Esterification

ANSWER / SOLUTION

Substances having ester as the functional group are formed by a reaction between a carboxylic acid and an alcohol. This reaction is known as esterification.

E.g. When ethanoic acid is heated with ethanol in a water bath in the presence of an acid catalyst (concentrated sulphuric acid), ethyl ethanoate and water are formed.



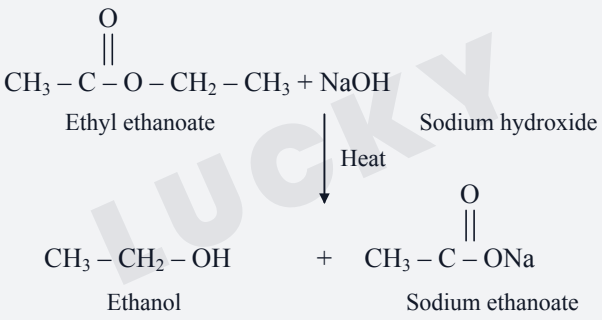
Q10 **Explain the following reaction with an example.**
Saponification

ANSWER / SOLUTION

When an ester is heated with dilute sodium hydroxide (an alkali) solution, it gets converted back to the original alcohol and sodium salt of the original carboxylic acid. Such a reaction is known as saponification.

$$\text{Ester} + \text{Sodium hydroxide} \xrightarrow{\text{Heat}} \text{Sodium carboxylate} + \text{Alcohol}$$

E.g. Ethyl ethanoate and sodium hydroxide react to give sodium ethanoate and ethanol.



Q11 **What are catalysts?**

TEXTBOOK **Exercise · Q.8 (g) · Page 134**

ANSWER / SOLUTION

Catalyst is a substance which increases the rate of a reaction without causing any disturbance to it.

Q12 **Give a chemical reaction in which a catalyst is used.**

TEXTBOOK **Exercise · Q.8 (g) · Page 134**

ANSWER / SOLUTION

E.g. In the hydrogenation of ethene, nickel or platinum is used as catalyst.

$$\underset{\text{Ethene}}{H_2C = CH_2} + H_2 \xrightarrow{\text{Ni/Pt}} \underset{\text{Ethane}}{CH_3\text{--}CH_3}$$

Q13 **Write the characteristics of ethanol.**

ANSWER / SOLUTION

1. It is colourless liquid at room temperature.
2. Its boiling point is $78\,^{\circ}C$.
3. It is soluble in water in all proportions.
4. Its aqueous solution is neutral.

LUCKY

Q14 **Write the answer as directed.**
What is meant by vinegar?

TEXTBOOK **Exercise · Q.8 (f) · Page 134**

ANSWER / SOLUTION

Vinegar is a type of liquid that consists of acetic acid (Ethanoic acid). It is basically produced by the process of fermentation of ethanol into ethanoic acid in the presence of bacteria.

Q15 **Write the uses of ethanol.**

ANSWER / SOLUTION

Ethanol being a good solvent, it is used in medicines such as tincture iodine (solution of iodine and ethanol), cough mixture and also in many tonics.

Q16 **Write the characteristics of ethanoic acid.**

ANSWER / SOLUTION

1. Ethanoic acid is a colourless liquid at ordinary temperature.
2. Its boiling point is $118\,^{\circ}\mathrm{C}$.
3. Melting point of pure ethanoic acid is $17\,^{\circ}\mathrm{C}$.
4. Its aqueous solution is acidic in nature and hence, turns blue litmus red.

10 Distinguish between

3 QUESTIONS ☐ Revised

Q1 **Distinguish between:**
Saturated hydrocarbons - Unsaturated hydrocarbons

ANSWER / SOLUTION

#	Saturated hydrocarbons	Unsaturated hydrocarbons
1.	Hydrocarbons having the carbon atoms linked to each other by single bonds are called saturated hydrocarbons.	Hydrocarbons having at least one carbon-carbon double or triple bond are called unsaturated hydrocarbons.
2.	On burning, saturated hydrocarbons give a clean, blue flame.	On burning, unsaturated hydrocarbons give a yellow flame with lots of black smoke.
3.	Saturated hydrocarbons do not decolourize bromine or iodine solutions.	Unsaturated hydrocarbons decolourize bromine or iodine solutions.
4.	Saturated hydrocarbons are less reactive.	Unsaturated hydrocarbons are usually highly reactive.

Q2 **Distinguish between:**
Open chain hydrocarbons - Closed chain hydrocarbons

ANSWER / SOLUTION

#	Open chain hydrocarbons	Closed chain hydrocarbons
1.	In open chain hydrocarbons, carbon atoms are linked to form a continuous straight or branched chain.	In closed chain hydrocarbons, carbon atoms are linked to form a ring structure.
2.	They are not aromatic compounds.	They can be aromatic compounds.
3.	**E.g.** Propane, Isobutane	**E.g.** Cyclohexane, Benzene

Q3 **Distinguish between:**
Alkane - Alkene

ANSWER / SOLUTION

#	Alkane	Alkene
1.	Alkanes are saturated hydrocarbons.	Alkenes are unsaturated hydrocarbons.
2.	They have C--C single bonds.	They have at least one $C = C$ double bond.
3.	They are characterized by the general formula is C_nH_{2n+2}.	They are characterized by the general formula is C_nH_{2n}.
4.	First member of homologous series is methane.	First member of homologous series is ethene.

11 Write short notes

5 QUESTIONS ☐ Revised

Q1 **Write a short note.**
Catenation power

ANSWER / SOLUTION

1. The unique property of carbon atom to form covalent bonds with other carbon atoms giving rise to extremely big molecules is called catenation power.
2. The covalent bond between two carbon atoms is strong and stable.
3. This is the reason carbon is bestowed with catenation power.
4. Due to catenation power, carbon can form a large number of compounds containing open chains or closed chains of carbon atoms.

Q2 **Write a short note.**
Characteristics of carbon

ANSWER / SOLUTION

1. **Catenation power:** Carbon has a unique ability to form strong covalent bonds with other carbon atoms; this results in the formation of big molecules. This property of carbon is called catenation power.
2. **Multiple bond formation:** Two carbon atoms can be bonded together by one, two or three covalent bonds. These are called a single bond, double bond, and triple bond respectively. Thus, carbon atom has the ability to form multiple bonds as well as single bonds.
3. **Tetravalency:** Being tetravalent one carbon atom can form bonds with four other atoms (carbon or any other). This results in formation of many compounds.
4. **Isomerism:** One more characteristic of carbon, which is responsible for large number of carbon compounds is isomerism.

Q3 **Write a short note.**
Functional groups in carbon compounds

ANSWER / SOLUTION

1. Many more types of carbon compounds are formed by formation of bonds of carbon with other elements such as halogens, oxygen, nitrogen, sulphur.
2. The atoms of these elements (hetero atoms) substitute one or more hydrogen atoms in the hydrocarbon chain and thereby the tetravalency of carbon is satisfied.
3. The compound acquire specific chemical properties due to these heteroatoms or the groups of atoms that contain heteroatoms, irrespective of the length and nature of the carbon chain in that compound.
4. Therefore, these hetero atoms or the groups of atoms containing hetero atoms are called functional groups.
   E.g. In CH_3--OH, --OH (alcohol) is the functional group-containing heteroatom 'oxygen'.

Q4 **Write a short note.**
Homologous series

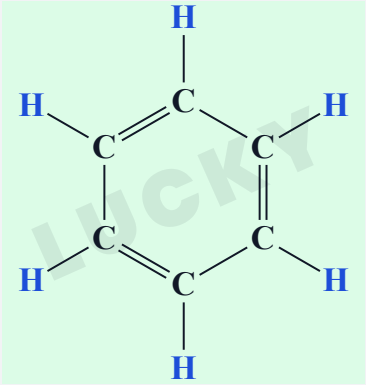
ANSWER / SOLUTION

1. The series of carbon compounds formed by joining the same functional group in the place of a particular hydrogen atom on the carbon chains having sequentially increasing length is called homologous series.
2. There are different homologous series in accordance with the functional group. For example, homologous series of alcohols, homologous series of carboxylic acids, homologous series of aldehydes, etc.
3. All the members of the homologous series are homologues of each other and they can be represented by the same general molecular formula.
4. Physical properties generally show a gradual change with the increase in molecular mass in the series whereas chemical properties of the members of the series show similarity because of the presence of the same functional group in them.

Q5 **Write a short note.**
Aromatic hydrocarbons

ANSWER / SOLUTION

1. Benzene is a cyclic unsaturated hydrocarbon. Its molecular formula is C_6H_6.
2. It has three alternate double bonds in its six-membered ring structure.
3. Structural formula of benzene:



4. Compounds having such characteristic unit in their structures are called aromatic compounds or aromatic hydrocarbons.

12 Answer the Following (Tables & Activities)

20 QUESTIONS ☐ Revised

Q1 **Complete the following table for the homologous series of alkanes.**

ANSWER / SOLUTION

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of $-CH_2-$ units	Boiling point °C
Methane	CH_4	CH_4	1	1	-162
Ethane	C_2H_6	CH_3--CH_3	2	2	-88.5
Propane	C_3H_8	CH_3--CH_2--CH_3	3	3	-42
Butane	C_4H_{10}	CH_3--CH_2--CH_2--CH_3	**4**	**4**	0
Pentane	C_5H_{12}	CH_3--CH_2--CH_2--CH_2--CH_3	**5**	**5**	36
Hexane	C_6H_{14}	CH_3--CH_2--CH_2--CH_2--CH_2--CH_3	**6**	**6**	69

Q2 **Complete the following table for homologous series of alcohols.**

ANSWER / SOLUTION

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of $-CH_2-$ units	Boiling point °C
Methanol	CH_4O	CH_3--OH	1	1	63
Ethanol	C_2H_6O	CH_3--CH_2--OH	2	2	78
Propanol	C_3H_8O	CH_3--CH_2--CH_2--OH	**3**	**3**	97
Butanol	$C_4H_{10}O$	CH_3--CH_2--CH_2--CH_2--OH	**4**	**4**	118

Q3 **Complete the following table for homologous series of Alkenes.**

ANSWER / SOLUTION

Name	Molecular formula	Condensed structural formula	Number of carbon atom	Number of $-CH_2-$ units	Boiling point $^\circ C$
Ethene	C_2H_4	$CH_2 = CH_2$	2	0	-102
Propene	C_3H_6	$CH_3\text{--}CH = CH_2$	3	1	-48
1-Butene	C_4H_8	$CH_3\text{--}CH_2\text{--}CH = CH_2$	**4**	**2**	-6.5
1-Pentene	C_5H_{10}	$CH_3\text{--}CH_2\text{--}CH_2\text{--}CH = CH_2$	**5**	**3**	30

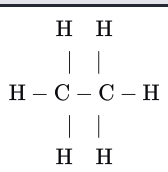
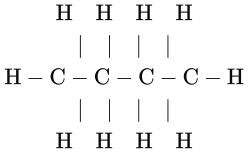
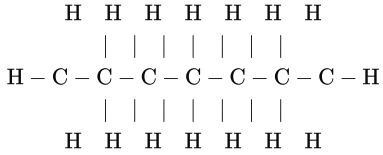
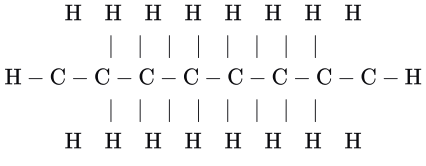
Q4 **Complete the given chart with writing the correct functional group of carbon compounds.**
(Ester, Aldehyde, Ketone, Carboxylic acid, Alcohol, Ether)

ANSWER / SOLUTION

Alcohol	--O--H
Aldehyde	$\begin{array}{c} O \\ \| \\ \text{--C--H} \end{array}$
Ketone	$\begin{array}{c} O \\ \| \\ \text{--C--} \end{array}$
Carboxylic acid	$\begin{array}{c} O \\ \| \\ \text{--C--O--H} \end{array}$
Ether	--O--
Ester	$\begin{array}{c} O \\ \| \\ \text{--C--O--} \end{array}$
Amine	$\begin{array}{c} \text{--N--H} \\ \vert \\ H \end{array}$

Q5 **Complete the following table with writing the correct structural formula and molecular formula.**

ANSWER / SOLUTION

Straight chain of carbon atoms	Structural formula	Molecular formula	Name
C-C		C_2H_6	Ethane
C-C-C-C		C_4H_{10}	Butane
C-C-C-C-C-C-C-		C_7H_{16}	**Heptane**
C-C-C-C-C-C-C-C		C_8H_{18}	**Octane**

Q6 **The table shows common names and structural formulae of a few carbon compounds. Complete the following table with writing IUPAC name of carbon compound.**

ANSWER / SOLUTION

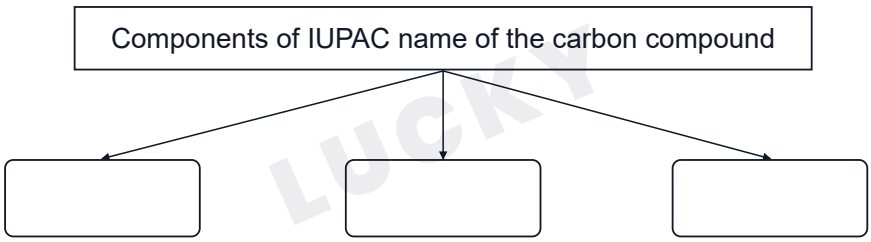
Sr. No.	Common name	Structural formula	IUPAC Name
1.	ethylene	$CH_2 = CH_2$	Ethene
2.	acetylene	$HC \equiv CH$	Ethyne
3.	acetic acid	CH_3--$COOH$	Ethanoic acid
4.	methyl alcohol	CH_3--OH	Methanol
5.	ethyl alcohol	CH_3--CH_2--OH	Ethanol
6.	acetaldehyde	CH_3--CHO	Ethanal
7.	acetone	CH_3--CO--CH_3	Propanone
8.	ethyl methyl ketone	CH_3--CO--CH_2--CH_3	Butanone
9.	ethyl amine	CH_3--CH_2--NH_2	Ethanamine
10.	n-propyl chloride	CH_3--CH_2--CH_2--Cl	1-Chloropropane

Q7 **Complete the following activity.**

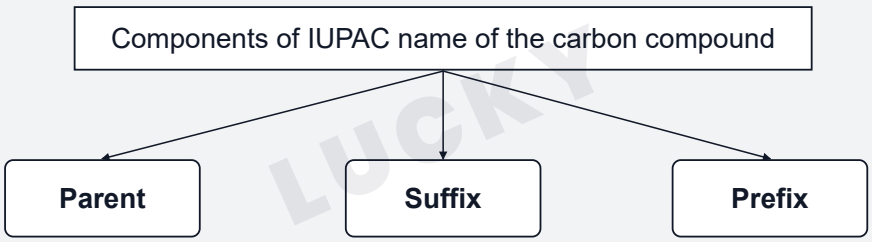
ANSWER / SOLUTION

Boiling point of ethanol	→	$78\,^\circ C$
General name of ethanol	→	**Ethyl alcohol or spirit**
Use of ethanol	→	**As a solvent in medicines**
Boiling point of ethanoic acid	→	$118\,^\circ C$
Melting point of pure ethanoic acid	→	$17\,^\circ C$

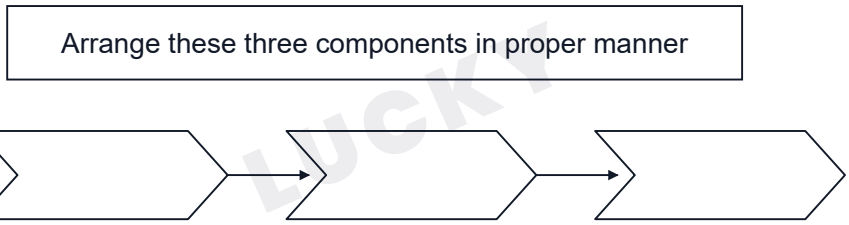
Q8 **Complete the following activity.**



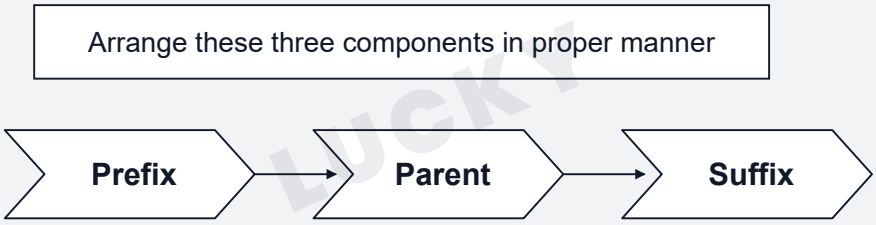
ANSWER / SOLUTION



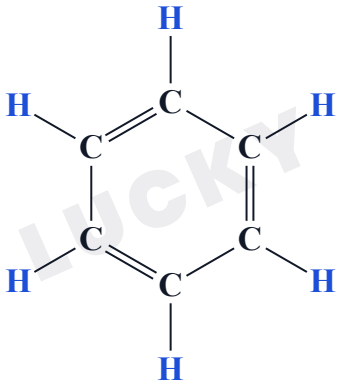
Q8. (b) **Complete the following activity.**



ANSWER / SOLUTION



Q9 **Observe the structural formula and answer the following questions.**



1. Write the name of the given hydrocarbon.
2. The given hydrocarbon is included in which type of hydrocarbon?
3. What is the kind of compounds with the above characteristic structure called?

ANSWER / SOLUTION

1. The given hydrocarbon is benzene.
2. The given hydrocarbon is included in cyclic hydrocarbon.
3. The compounds having this characteristic unit in their structure are called aromatic compounds.

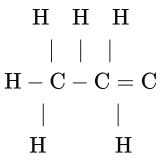
Q10 **Complete the following chart by using examples given in brackets.**
(isobutylene, cyclohexane, propane, cyclohexene, cyclopentane, benzene, propyne, isobutane, propene)

ANSWER / SOLUTION

Straight chain hydrocarbons	Branched chain hydrocarbons	Cyclic hydrocarbons
Propane	Isobutylene	Cyclohexane
Propyne	Isobutane	Cyclohexene
Propene		Cyclopentane
		Benzene

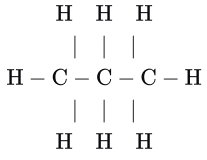
Q12 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Propene**

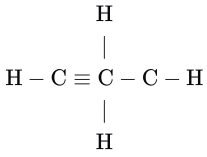
Q13 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Propane**

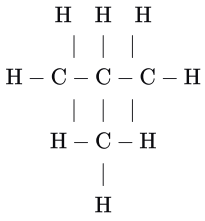
Q14 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Propyne**

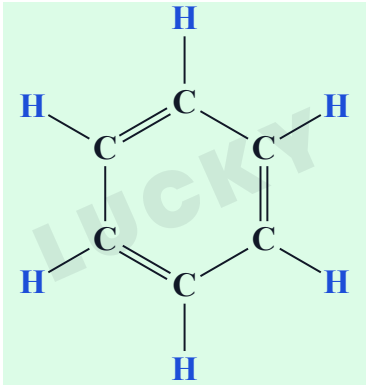
Q15 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Isobutane**

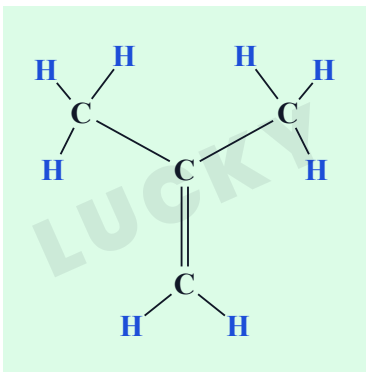
Q16 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Benzene**

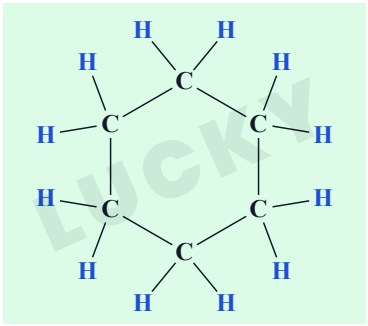
Q17 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Isobutylene**

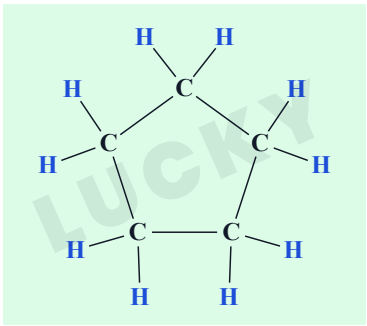
Q18 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Cyclohexane**

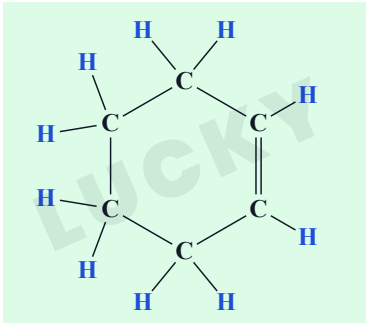
Q19 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

ANSWER / SOLUTION

	**Cyclopentane**

Q20 **Complete the following activity.**
Write the names of the hydrocarbons for the following structural formula.

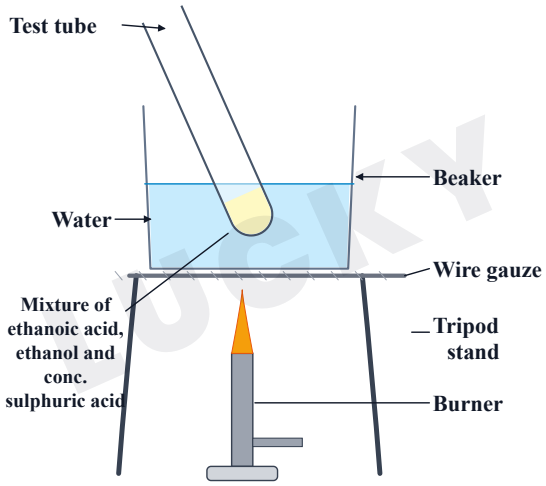
ANSWER / SOLUTION

	**Cyclohexene**

13 Answer the following questions

2 QUESTIONS ☐ Revised

Q1 **Observe the figure and write the answers to the following questions.**



1. Write the name of the reaction shown in the following figure.
2. Write the above chemical reaction in the form of a balanced equation.
3. Write the name of the product produced in the above reaction, write a use.
4. Write the name of the catalyst used in the above reaction.

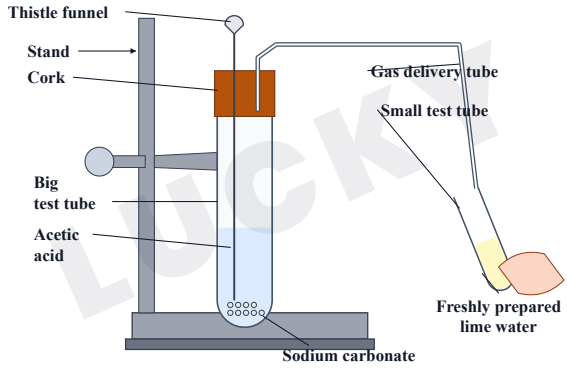
ANSWER / SOLUTION

1. The reaction shown in the figure is esterification.
2. Balanced equation of the reaction between ethanol and ethanoic acid is as follows:

$$C_2H_5OH + CH_3COOH \xrightarrow[\text{Heat}]{\text{Conc. } H_2SO_4} CH_3COOC_2H_5 + H_2O$$

3. The product formed is ethyl ethanoate.
   Use: It is used for making fragrances and flavouring agents.
4. The catalyst used in the reaction is concentrated sulphuric acid.

Q2 **Write the answers to the questions by observing the following figure.**



1. Write the chemical reaction shown in the figure above in the form of a balanced equation.
2. Write the name of the gas coming out of the large test tube in the above chemical reaction.
3. Why do small bubbles appear in the small test tube?
4. What is the change in colour of lime water?

ANSWER / SOLUTION

1. Balanced equation of the reaction between acetic acid and sodium carbonate is as follows:
   $2CH_3COOH_{(aq)} + Na_2CO_{3(s)} \rightarrow 2CH_3COONa_{(aq)} + H_2O_{(l)} + CO_{2(g)} \uparrow$
2. The gas formed in the large test tube is carbon dioxide (CO_2) gas.
3. The carbon dioxide gas produced in big test tube passes through bent gas delivery tube to the small test tube containing lime water. Hence, bubbles are seen in the small test tube.
4. Lime water turns milky.

Keep revising. Keep winning.
Tesla Coaching Classes, Kawarabandh • Founder: Jitendra Yele • Tesla Result System™